RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
Bengaluru - 560059, Karnataka, India

# Department of Mathematics

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| Date: 16/04/2026 | CIE - 1 | Max. Marks:100 |
| Semester: VI | UG | Duration: 2 Hrs. (2.30pm-4.30pm) |
| Course Title: Mathematical Modelling | | Course Code: MA266TEU |

| SL.No | Quiz Questions | M | BT | CO |
|---|---|---|---|---|
| 01 | The maximum horizontal distance covered by a body of mass 5 kg is projected from the ground at an angle of 45° with an initial velocity of 20 m/s is _____. (Take acceleration due to gravity as $g \approx 9.8\ m/s^2$) | 2 | L1 | 1 |
| 02 | The growth in the number of software engineers in a country follows the law $\frac{dx}{dt} = \frac{1}{20}x$. If initially there were 800 engineers, then after 20 years, their approximate number will be _____ and their doubling period is approximately _____ years. | 2 | L1 | 2 |
| 03 | A heated metal sphere at 120°C is placed in a room maintained at 30°C. If the cooling constant is $k = 0.03$, then the temperature of the sphere after 15 minutes is _____. | 2 | L2 | 3 |
| 04 | The sequence 2, 5, 10, 17, 26, ... can be modelled by a difference equation of the form _____. | 2 | L2 | 2 |
| 05 | State Fick's law of diffusion model. | 2 | L1 | 1 |

| SL.No | Test Questions | M | BT | CO |
|---|---|---|---|---|
| 1a. | Model the motion of a projectile launched in a vacuum with an initial velocity at a given horizontal angle. Formulate the governing equations of motion, solve them and determine key characteristics such as time of flight, maximum height and horizontal range. | 6 | L3 | 3 |
| 1b. | Discuss how mathematical modelling is classified. | 4 | L1 | 2 |
| 2. | A nutrient is periodically injected into a patient's bloodstream at fixed intervals of time $T$. Meanwhile, the nutrient is continuously absorbed by the body at a rate proportional to its concentration $c(t)$ in the blood at time $t$. Derive the differential equation governing $c(t)$. Solve it and interpret the long-term behaviour of the nutrient concentration in the bloodstream. | 10 | L4 | 4 |
| 3. | Develop a differential equation model based on the logistic law of population growth, solve it and analyse any one real life scenario. | 10 | L3 | 2 |
| 4 | Describe the systematic 12-step procedure for tackling a problem through the use of mathematical modelling. | 10 | L1 | 1 |
| 5a | Solve the difference equation $x_{n+1} - 0.3x_n - 6 = 0,\ x_0 = 10, n \geq 0$. Hence Find the limiting (convergent) value of the sequence. | 5 | L3 | 3 |
| 5b. | Form the difference equation for the sequence 0, 1, 1, 2, 3, ... and solve it. | 5 | L2 | 2 |

Blooms Taxonomy, CO-Course Outcomes, M-Marks

| | Particulars | | CO1 | CO2 | CO3 | CO4 | L1 | L2 | L3 | L4 | L5 | L6 |
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| Marks Distribution | Quiz | Max Marks | 04 | 4 | 2 | - | 04 | 4 | 2 | | | |